

Bluestock Mutual Fund Analytics Capstone

1. Executive summary

The Bluestock Mutual Fund Analytics Capstone is an end-to-end data analytics project designed to analyze the performance and risk characteristics of mutual fund schemes using historical market data. The project integrates multiple mutual fund datasets into a structured analytics pipeline that automates data extraction, transformation, validation, storage, analysis, and visualization. Cleaned data is stored in a SQLite database and processed using Python-based analytical workflows. Key investment metrics such as CAGR, volatility, Sharpe ratio, maximum drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR) are calculated to evaluate fund performance comprehensively. An interactive Power BI dashboard enables investors and analysts to compare schemes, assess risk-adjusted returns, and identify category-level trends. The project demonstrates practical applications of ETL engineering, database design, financial analytics, and business intelligence. It provides a scalable framework that can be extended for automated portfolio monitoring and investment research.

2. Project objectives

The primary objective of this project is to build a robust analytics platform for mutual fund evaluation and investment decision support. The ETL process is designed to automatically clean, standardize, and validate data collected from multiple financial sources. A centralized SQLite database is created to ensure efficient storage and retrieval of analytical data. Exploratory Data Analysis (EDA) is performed to understand fund distributions, category patterns, missing values, and return behavior. Financial performance metrics including CAGR, annualized volatility, Sharpe ratio, drawdown, VaR, and CVaR are computed for all eligible mutual fund schemes. The project also aims to develop an interactive dashboard that allows dynamic filtering and comparison across categories and funds. Finally, the analysis seeks to generate actionable insights for investors by emphasizing risk-adjusted performance rather than return alone.

3. Data sources

The analysis is based on multiple datasets representing different aspects of the Indian mutual fund industry. The AMFI fund master dataset provides scheme-level metadata such as AMFI code, fund name, category, and fund house. Historical NAV data serves as the primary time-series input for return calculations and risk measurement. AUM datasets provide information about fund size and market participation. SIP inflow and category inflow datasets help evaluate investor behavior and capital allocation trends. Folio count data is used to measure retail investor participation across categories. Scheme performance datasets provide benchmark and historical return comparisons. Portfolio holdings datasets offer insights into asset allocation and diversification patterns. Benchmark indices such as NIFTY 50 and NIFTY 100 are used for comparative performance analysis and alpha/beta estimation.

4. ETL design

The ETL pipeline is implemented using Python with Pandas and SQLAlchemy to automate data processing. Raw CSV files are first ingested from multiple source directories and standardized into a common schema. Column names are normalized by removing spaces, converting text to lowercase, and replacing special characters. Date fields are converted into consistent datetime formats, and invalid records are identified through validation checks. Duplicate entries are removed to prevent multiple NAV observations for the same scheme and date. Missing NAV values are handled through forward-filling and business-day reindexing to maintain continuous time-series data. AMFI codes are validated against the fund master dataset to ensure referential integrity. Cleaned datasets are exported to processed CSV files and simultaneously loaded into SQLite tables.

5. Database architecture

A relational SQLite database is designed to support efficient analytical queries and reporting workflows. The `fund_master` table stores static scheme-level information including AMFI code, scheme name, category, and fund house. The `nav_history` table contains historical NAV observations indexed by AMFI code and date. A one-to-many relationship exists between `fund_master` and `nav_history`, where a single scheme can have thousands of daily NAV records. Primary and foreign key relationships ensure data consistency across tables. Indexing on AMFI code and date significantly improves query performance for time-series analysis. The database structure supports aggregation, category-level comparisons, rolling return calculations, and dashboard integration. SQLite is selected because of its lightweight architecture, portability, and seamless integration with Python analytical libraries.

6. EDA findings

Exploratory Data Analysis reveals several important characteristics of the mutual fund dataset. Category distribution analysis shows that equity funds dominate the dataset, followed by debt, hybrid, and other categories. NAV distributions are positively skewed, indicating that a relatively small number of schemes have accumulated substantially higher NAV values over time. Missing-value analysis confirms that most data quality issues originate from non-trading days and inconsistent historical reporting. Daily return distributions exhibit near-normal behavior with noticeable fat tails, reflecting occasional market shocks. Volatility varies significantly across categories, with small-cap and thematic funds generally displaying higher return dispersion. Fund concentration analysis indicates that a limited number of large AUM schemes account for a substantial share of industry assets. These findings justify the use of risk-adjusted performance metrics in subsequent analysis.

7. Performance analysis

Performance evaluation is conducted using standardized financial analytics methodologies. Compound Annual Growth Rate (CAGR) measures long-term wealth creation across different mutual fund schemes. Annualized volatility quantifies return variability and overall investment risk. The Sharpe ratio evaluates excess return generated per unit of risk, enabling comparison across funds with different volatility levels. Maximum drawdown identifies the largest historical decline from peak NAV, providing insight into downside exposure. Historical Value at Risk (VaR 95%) estimates the potential one-day loss under adverse market conditions. Conditional Value at Risk (CVaR 95%) measures the average loss beyond the VaR threshold and captures tail-risk severity. These metrics collectively provide a comprehensive assessment of return, risk, and downside protection.

8. Dashboard overview

The Tableau dashboard is designed to present analytical insights through an intuitive and interactive interface. The Executive Overview page summarizes total schemes, category distribution, average CAGR, average Sharpe ratio, and overall market trends. The Performance Analytics page enables fund-level comparison of CAGR, rolling returns, cumulative growth, and benchmark-relative performance. The Risk Analytics page visualizes volatility, drawdown, VaR, and CVaR through comparative charts and ranking tables. The Fund Recommender page highlights top-performing schemes based on configurable filters such as category, risk level, and Sharpe ratio thresholds. Interactive slicers allow users to dynamically explore categories, fund houses, and time periods. Drill-through functionality enables detailed scheme-level analysis from high-level summaries. The dashboard supports investment screening and portfolio research in a user-friendly format.

9. Key findings

The analysis demonstrates that higher returns do not necessarily correspond to superior investment quality. Several funds with exceptional CAGR also exhibit very high volatility and poor drawdown characteristics. Risk-adjusted measures such as the Sharpe ratio provide a more balanced evaluation of performance than return metrics alone. Maximum drawdown varies substantially across categories, with small-cap and sectoral funds experiencing significantly deeper historical declines. VaR and CVaR analysis reveals meaningful differences in downside tail risk across schemes. Large diversified equity funds generally provide more stable risk-adjusted performance than concentrated thematic funds. Category diversification substantially reduces portfolio-level risk exposure. These findings emphasize the importance of combining return and risk metrics when selecting mutual fund investments.

10. Limitations

The project has several practical limitations that should be considered when interpreting the results. Historical performance metrics are backward-looking and cannot guarantee future investment outcomes. Benchmark alignment may affect the accuracy of alpha and beta estimates for schemes with changing investment mandates. Some datasets contain incomplete historical records, requiring interpolation and data-cleaning assumptions. The analysis does not explicitly incorporate taxation, expense ratio changes, exit loads, or investor-specific cash-flow timing. Macroeconomic factors such as interest rate cycles, inflation, and policy changes are not modeled directly. The SQLite database is suitable for analytical workloads but may require migration to PostgreSQL or cloud-based systems for large-scale production deployment. Results should therefore be interpreted as analytical indicators rather than investment recommendations.

11. Recommendations

Investors should prioritize risk-adjusted performance metrics when comparing mutual fund schemes. The Sharpe ratio provides a more informative measure of investment efficiency than CAGR alone. Portfolio diversification across equity, debt, and hybrid categories can significantly reduce drawdown risk. Regular monitoring of maximum drawdown helps investors evaluate downside tolerance and capital preservation. VaR and CVaR should be incorporated into periodic portfolio risk assessments, especially during volatile market conditions. The ETL pipeline should be automated through scheduled data refreshes to maintain current analytical outputs. Benchmark comparisons should be updated periodically to improve performance attribution accuracy. Investors should combine quantitative analytics with qualitative assessment of fund management, investment strategy, and market conditions.

12. Future work

Several enhancements can significantly improve the analytical capabilities of this platform. Monte Carlo simulation can be added to estimate probabilistic future return distributions and portfolio outcomes. Markowitz portfolio optimization can be implemented to construct efficient portfolios under different risk-return objectives. A Streamlit web application can replace desktop dashboards to enable cloud-based interactive access. Automated NAV refresh pipelines can be scheduled using Airflow or cron-based workflows. Email reporting can be integrated to distribute periodic performance summaries and risk alerts. Machine learning models can be developed for return forecasting and anomaly detection. Cloud deployment using PostgreSQL and Power BI Service can improve scalability and collaboration. These enhancements would transform the project into a production-grade mutual fund analytics platform suitable for professional investment research.